

BUS713 Financial Analytics: Workshop 1

Introduction to R for Financial Analytics

This workshop teaches R as a tool for financial analysis. Every task builds toward one outcome: downloading real market data, calculating returns, and interpreting what they tell you. Work through each problem yourself, in order; each task assumes you completed the one before it.

Setup

Install R, then RStudio, from <https://posit.co/download/rstudio-desktop/>, accepting default options. RStudio is the interface; R is the engine underneath it. Open RStudio and locate its four panes: Source (top-left, write scripts), Console (bottom-left, run code), Environment (top-right, stored objects), and Plots/Files/Help (bottom-right).

Install and load today's packages (tidyverse, tidyquant, tsibble, forecast, fredr) before starting Task 1.

Task 1: R Fundamentals

Scenario: You've joined an investment bank's research team. Your manager wants a briefing on US consumption and unemployment using the built-in economics dataset (monthly US data since 1967, part of ggplot2).

Load and inspect the economics dataset. R follows BEDMAS. mean(), sd(), min(), max() are the core descriptive statistics; log() is the natural logarithm, central to return calculations later.

1. Predict, then check: $2 + 3 * 4$, $(2 + 3) * 4$, and $10 / (2 - 3)$. A colleague writes a compound return as $1000 * 1 + 0.08 ^ 3 - 1000$ instead of $1000 * (1 + 0.08) ^ 3 - 1000$. Run both and explain the gap using BEDMAS.
2. Calculate the mean and standard deviation of unemploy for 2000-2007 and separately for 2009-2015. What does the change in standard deviation say about the labour market before versus after the GFC?
3. Calculate log(pce) for the latest observation and the observation 12 months earlier. Compare the log difference to the arithmetic percentage change $(pce_end - pce_start) / pce_start$. How close are they?

Task 2: Wrangling Data with dplyr

Scenario: Your manager wants specific answers, not raw data dumps: which months looked weakest, and a reusable way to measure change.

filter() keeps rows meeting a condition. group_by() splits data into groups before summarise() collapses each group to one row. mutate() adds a new column.

4. Count how many months since 2000 have both pce below its own median and unemploy above 10,000 (wrap your filter() in nrow()). Then loosen it to either condition (use | instead of &). How much does the count change?
5. Write a function, pct_change(old_value, new_value), that returns the percentage change between two numbers. Use it to find the percentage change in unemploy and in pce between the first and last row of economics. Write a second function, annualise_change(), that converts a pct_change result over n years to a compound annual rate: $(1 + x)^{(1/n)} - 1$.
6. Create a decade column with mutate(), then use group_by() and summarise() to calculate average unemploy and average pce by decade. Which decade had the highest average unemployment?

Task 3: Visual Storytelling with ggplot2

Scenario: Numbers get skimmed; a labelled chart for the investment committee gets remembered.

Every ggplot2 chart follows the same recipe: ggplot(data, aes(...)) + geom_...() + labs(...) + theme_...().

7. Build a line chart of unemploy over time with geom_line(), labelled with labs(). Rebuild it for pce. Add a dashed red vertical line at the 2008 recession start using geom_vline().
8. Build a scatter plot of pce against unemploy with geom_point(). Colour the points by decade (create the decade column first). Does the relationship look consistent across decades?
9. Combine unemploy and pce into one long data frame with a series label column (rbind two mutated subsets), then chart both with facet_wrap(~ series, scales = "free_y"). Remove scales = "free_y" and explain what breaks.

Task 4: Real Stock Data and Returns

Scenario: You're an equity analyst comparing Coca-Cola (KO) and PepsiCo (PEP) for a portfolio review. Use the adjusted price throughout; it accounts for splits and dividends.

Pull each stock's price history with `tq_get()`. Arithmetic return: $(P_{\text{end}} - P_{\text{start}}) / P_{\text{start}}$. Log return: $\log(P_{\text{end}} / P_{\text{start}})$. Log returns are additive across time; arithmetic returns are not.

10. Download KO and PEP price history from 2000-01-01 using `tq_get()`. Rebase both to start at 100 (hint: divide by the first adjusted price and multiply by 100) and chart them together to compare growth, not dollar level.
11. Calculate monthly arithmetic and monthly log returns for KO and for PEP using `tq_transmute()` with `periodReturn()`. Repeat with `period = "yearly"` for both stocks.
12. Pick one month and calculate the dollar-based gap between its arithmetic and log return by hand. In which market conditions would you expect this gap to widen?

Task 5: CAGR and Recommendation

Scenario: Your manager wants a one-page growth comparison of KO and PEP before the investment committee meets.

CAGR: $(\text{end price} / \text{start price})^{(1 / \text{years})} - 1$, where years is the number of days between the first and last date divided by 365.25.

13. Calculate CAGR for both KO and PEP. Compare each CAGR to the average annual arithmetic return from Task 4. Then remove the outer brackets from the formula and explain the resulting error using BEDMAS.
14. Write a 3 to 4 sentence memo to the investment committee recommending KO or PEP based on growth, citing each stock's CAGR.

Wrap-Up

You installed R and RStudio, used core dplyr and base R tools on real economic data, built labelled ggplot2 charts, and downloaded real stock data to calculate arithmetic returns, log returns, and CAGR. Keep your KO and PEP scripts; next session builds directly on them.